Big ideas.
Real examples.
Zero boring.

GRADE 8 ICT & DESIGN · TERM 1

HOW TO REALLY GET THE DESIGN PROCESS

Your Study Guide — Version 2 ✏️

From empathy to app — everything you need to know, made easy to actually read.

How to Use This Guide 📖

This guide isn't organized week by week — it's organized by big idea, so you can jump to any concept from the whole unit and it'll make sense on its own. Every chapter follows the same 5-part pattern:

1. **Big Idea** — The one sentence that matters most.
2. **Key Terms** — The vocabulary you need, in plain language.
3. **Did You Know?** — A real-world connection to make it stick.
4. **Try This** — A quick activity to practice the idea yourself.
5. **Quick Check** — Questions to test if it really sunk in.

MEET TEAM STUDYBUDDY

Throughout this guide we'll follow one running example: a team of students designing a homework-help app called StudyBuddy, from their first real conversation with a user all the way to their final presentation. It's a made-up team — but every step is exactly the process you're using in your own project.

What's Inside ✨

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2. Define & Prepare
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7. Building & Testing Your Paper Prototype
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CHAPTER 1

Empathy & Understanding Your User

Great design doesn't start with an idea. It starts with a person.

Meet **Team StudyBuddy** — a group of students who spent this whole term designing a real app, from a blank page to a working prototype. We'll follow their journey through every chapter of this guide.

Before Team StudyBuddy wrote a single line of code, they didn't ask "what should our app do?" They asked "who needs help, and with what?" They talked to classmates who said homework felt overwhelming and hard to track. That one conversation shaped everything that came next — because **empathy** means understanding what someone else actually needs, not what you'd guess they need.

KEY TERMS

Empathy
Understanding and sharing another person's feelings or perspective.

User
The person a product or design is actually made for.

Need
Something a user requires from a product for it to work for them.

User Profile
A description of a person's age, abilities, habits, and needs.

Perspective
Seeing a situation from someone else's point of view.

Inclusive
Designed to work well for as many different people as possible.

Accessibility
How easily a product can be used by people with different abilities.

Evidence
Specific facts or details that support a design decision.

DID YOU KNOW?

The OXO Good Grips kitchen tools were originally designed for people with arthritis, who needed easy-to-grip handles. Turns out — everyone loved them. Designing for one specific user's real need often makes a product better for everyone.

TRY THIS

Ask someone at home about one small thing that annoys them every day. Write down their need in one sentence — not your guess, their actual words.

QUICK CHECK

- [ ] What's the difference between a "need" and a "want"?
- [ ] Why might designing for a specific user actually make a product better for more people?

CHAPTER 2

Define & Prepare

You can't solve a problem you haven't clearly named.

Team StudyBuddy had a pile of sticky notes full of possible needs — reminders, motivation, finding study partners, tracking grades. Instead of trying to fix everything, they grouped similar ideas into categories, then picked ONE **focus need**: "Students forget what homework is due and when."

That focus became their design criteria: whatever they built had to show upcoming homework clearly, in one place.

KEY TERMS

Define
Clearly stating what problem you're solving and for whom.

Brainstorm
Generating as many ideas as possible without judging them yet.

Category
A group of related ideas, organized to be easier to think about.

Criteria
The specific standards a solution needs to meet to succeed.

DID YOU KNOW?

In real brainstorming sessions, the rule is: no judging ideas yet. Research shows judging ideas too early kills good ones before they've had a chance to develop.

TRY THIS

Pick a problem you actually have this week. Brainstorm 5 directions without judging any of them, then pick your favorite.

QUICK CHECK

- [ ] What's the difference between brainstorming and defining?
- [ ] Why sort sticky notes into categories before choosing a focus need?

CHAPTER 3

Prototyping Your First Idea

The fastest way to a good idea is a lot of rough ones.

Team StudyBuddy didn't wait until they had the "perfect" app idea. They sketched a rough, quick version — a **prototype** — of a homework tracker screen, labeled which parts solved their focus need, and showed it to classmates.

The first version wasn't good. That was the point: it was fast to make and fast to learn from.

KEY TERMS

Prototype
An early, rough version of a design built to test an idea.

Iteration
Making a design better through repeated rounds of improving it.

Feedback
Specific, useful comments about how well something is working.

Design Rationale
The reasons behind a design decision — the WHY, not the WHAT.

DID YOU KNOW?

A famous design firm once had teams build a shopping-cart prototype out of foam, wire, and tape in one afternoon — testing a rough idea fast beats planning a perfect one slowly.

TRY THIS

Sketch a rough "prototype" of anything in under 5 minutes. Don't erase mistakes — just get the idea down.

QUICK CHECK

- [ ] Why would a designer intentionally make a "rough," not-perfect prototype?
- [ ] What's the difference between feedback and design rationale?

CHAPTER 4

User Interfaces & Usability

If people can't figure out how to use it, it doesn't matter how good the idea is.

When Team StudyBuddy tested their paper prototype, one partner acted as the "app" while another tried to complete a real task — find tomorrow's homework. The tester got stuck: too many buttons, no clear way back.

That's a **usability** problem, not a bad idea — the interface just needed to be clearer.

KEY TERMS

User Interface
The screens, buttons, and controls a person sees and touches.

Usability
How easy and pleasant something is to actually use.

Navigation
How a user moves between different screens of an app.

Screen
One individual view or page within an app.

DID YOU KNOW?

Good navigation usually means a user should never wonder "where am I?" or "how do I get back?" If they have to guess, the interface needs work — not the user.

TRY THIS

Open any app you use often. Try to find a feature you rarely use. Notice how many taps it takes, and whether you ever feel lost.

QUICK CHECK

- [ ] What's the difference between a good idea and a usable interface?
- [ ] Name one part of a UI that helps with navigation.

CHAPTER 5

Using Feedback to Improve

Feedback isn't criticism — it's data.

After testing, Team StudyBuddy had a pile of sticky-note feedback: "confusing button," "love the color," "missing a way to mark homework done."

They sorted it into categories, found the most common issue, and **prioritized** fixing that one first instead of trying to fix everything at once.

KEY TERMS

Prioritize
Deciding what to work on first based on what matters most.

Revise
Changing something to make it better, based on feedback.

I Like / I Wish / I Wonder
A feedback protocol that keeps critique specific and kind.

DID YOU KNOW?

"I Like / I Wish / I Wonder" is used far beyond classrooms — real product teams at tech companies use almost the exact same structure to keep feedback specific and kind.

TRY THIS

Next time a friend shows you something they made, respond with one "I like," one "I wish," and one "I wonder" instead of just "looks good."

QUICK CHECK

- [ ] Why sort feedback into categories before acting on it?
- [ ] What does it mean to "prioritize" feedback?

CHAPTER 6

Finding Real User Needs Through Interviews

The best app ideas come from listening, not guessing.

Team StudyBuddy read through interview notes from real students. One line stood out: "I always remember at 9pm that I have math homework due tomorrow."

That single sentence was evidence of a **barrier** (forgetting) and an **opportunity** (a reminder that actually gets noticed) — and it became the seed of their whole app idea.

KEY TERMS

Interview
Asking someone questions directly to learn about their needs.

Barrier
Something that gets in the way of a user doing what they need.

Opportunity
A chance to solve a problem or make something better.

Insight
A useful realization about a user that explains why they need something.

App Idea
A specific concept for an app that solves a real need.

DID YOU KNOW?

Just guessing what someone needs is risky — studies on failed products consistently point back to teams who skipped talking to real users.

TRY THIS

Ask someone a genuinely open question — "what's the most annoying part of your day?" — and just listen. Write down their answer word-for-word.

QUICK CHECK

- [ ] What's the difference between a barrier and an opportunity?
- [ ] Why does an app idea need to be linked to a specific need?

CHAPTER 7

Building & Testing Your Paper Prototype

Test it before you build it — paper is cheaper to fix than code.

Team StudyBuddy built every screen of their app idea out of paper before touching a computer. They mapped the **screen flow** — how a user moves from screen to screen — then swapped prototypes with another team and watched someone else try to use it.

That's when they discovered their "Add Homework" button was impossible to find.

KEY TERMS

Paper Prototype
A version of an app made entirely out of paper, used to test an idea.

Screen Flow
The path a user follows moving from one screen to the next.

Test
Having a real person try to use your design so you can see what works.

Stakeholder
Anyone who has an interest in or is affected by a project.

DID YOU KNOW?

Paper prototyping is a real, professional practice — many app and game studios still test ideas on paper first because it's fast and cheap to change.

TRY THIS

Draw 3 connected screens of an app idea on paper. Add arrows showing how a user moves between them.

QUICK CHECK

- [ ] Why test a paper prototype before building anything digital?
- [ ] Who is a "stakeholder" in a school app project?

CHAPTER 8

Apps for Good & Market Research

Before you build, check what already exists — and ask who it's really for.

Team StudyBuddy looked at existing homework apps. Some were cluttered with ads, some required a paid account, some didn't work offline.

That research didn't discourage them — it showed exactly what their app could do differently, and reminded them to think about who might be left out.

KEY TERMS

Apps for Good
Apps designed to help solve a real problem, not just for profit.

Responsible
Making design choices that consider the wellbeing of users.

Market Research
Looking at what already exists to see what's missing.

Competitor
An existing app or product that solves a similar problem.

Ethical
Doing what's right and fair, especially with user information.

DID YOU KNOW?

"Apps for good" is a real, growing category of technology built specifically to solve social or community problems, not just to make money.

TRY THIS

Pick an app you use. Name one competitor app that's similar, and one thing your app does better.

QUICK CHECK

- [ ] Why research competitor apps before designing your own?
- [ ] What makes an app idea "responsible," not just "cool"?

CHAPTER 9

UI Elements in Your App

Every button, box, and icon on a screen should earn its place.

When Team StudyBuddy planned their screens, they didn't just add elements randomly. Each button, text input, and icon had a job — and they used **visual hierarchy** to make sure the most important thing was noticed first.

KEY TERMS

Button
A UI element a user taps or clicks to trigger an action.

Text Input
A UI element where a user types information into an app.

Dropdown
Shows a list of options when tapped, letting a user pick one.

Icon
A small picture or symbol representing an action or piece of information.

Visual Hierarchy
Arranging elements so the most important thing is noticed first.

DID YOU KNOW?

Designers often test visual hierarchy by squinting at a screen — if you can still tell what's most important when it's blurry, the hierarchy is working.

TRY THIS

Look at any app screen. Which element is biggest or boldest? Is that the thing you're supposed to notice first?

QUICK CHECK

- [ ] Name a UI element you'd use to let a user pick from a list.
- [ ] What is “visual hierarchy” for?

CHAPTER 10

From Paper to Digital: App Lab

This is where your paper prototype becomes a real, working screen.

With a tested paper prototype in hand, Team StudyBuddy opened Code.org's App Lab and used **Design Mode** to recreate their screens for real — dragging in buttons and text, matching what they'd already tested on paper.

Nothing here was a guess anymore; it was already user-tested.

KEY TERMS

App Lab
The Code.org tool used to build real, working apps.

Design Mode
The part of App Lab where you build a screen's visuals before adding code.

DID YOU KNOW?

Design Mode in App Lab is deliberately separate from the code — you build what a screen looks like first, then add what happens when someone taps something.

TRY THIS

Before your next App Lab session, sketch on paper exactly what you plan to build digitally. Compare the two when done.

QUICK CHECK

- [ ] Why build a paper prototype before opening App Lab?
- [ ] What does Design Mode let you build?

CHAPTER 11

Events, Debugging & Presenting Your App

A working app is screens connected by events — and a few bugs along the way.

Team StudyBuddy linked their screens with events — tapping a button now actually took you to the next screen. Along the way, things broke. That's **debugging** — finding and fixing what's not working yet, not a sign something went wrong.

By the final week, they didn't just show their app — they presented their whole journey: the user need, feedback, bugs fixed, and what they'd still improve.

KEY TERMS

Event
Something a user does that tells the app to respond.

Link (Screens)
Connecting one screen to another so tapping moves the user.

Debug
Finding and fixing something that isn't working correctly.

Bug
Something in your app that doesn't work the way it's supposed to.

Feature
Something your app does or offers, whether finished yet or not.

Presentation
Sharing your work and explaining your process to an audience.

Reflection
Thinking back on what worked, what didn't, and what you'd change.

DID YOU KNOW?

The word "bug" in computing dates back to a real moth found stuck in an early computer in 1947 — and so did the practice of calmly hunting down what's broken.

TRY THIS

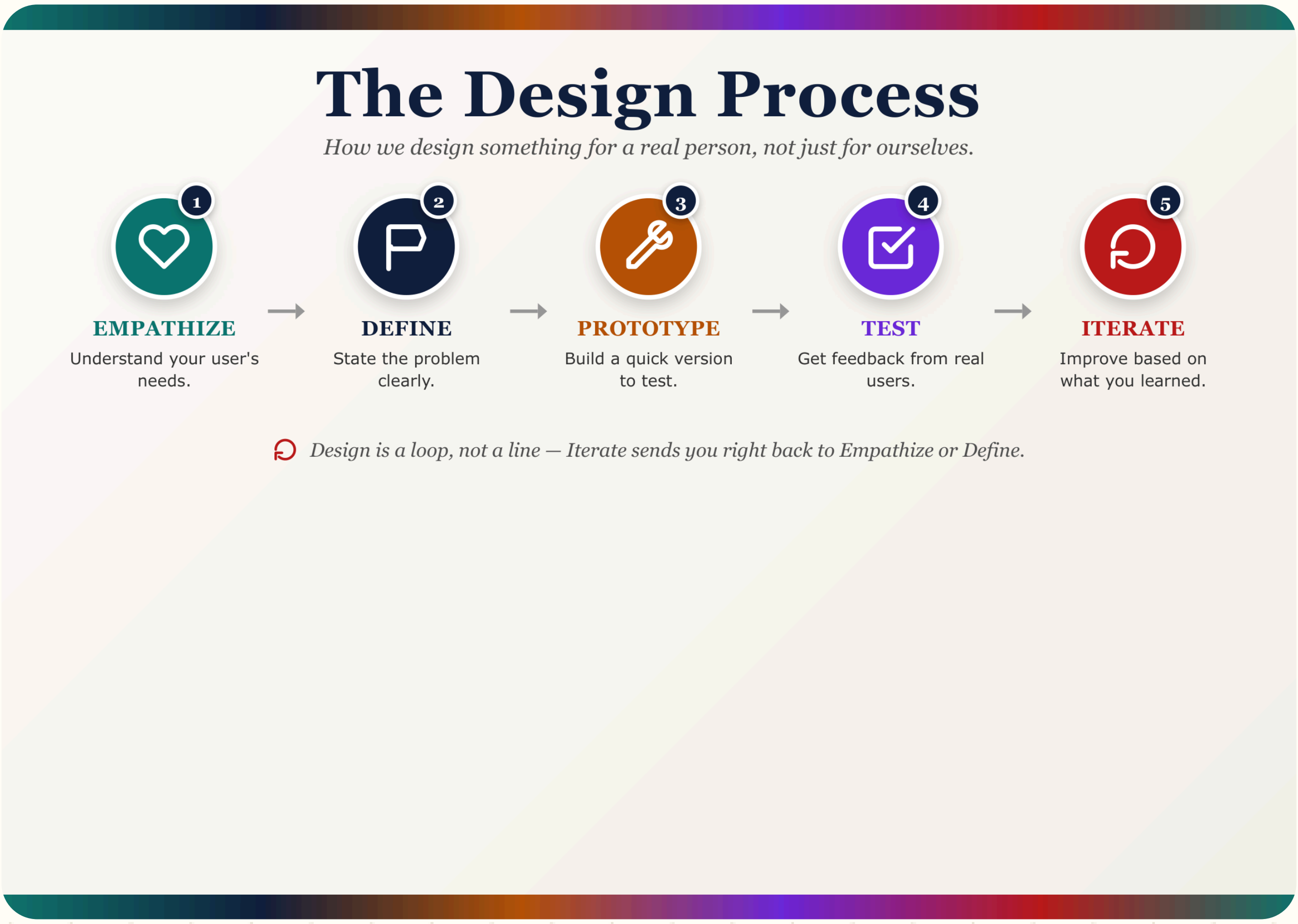
Next time something you make doesn't work right away, call it a "bug to debug" instead of a failure.

QUICK CHECK

- [ ] What's the difference between a bug and a feature?
- [ ] Why does a good presentation include what DIDN'T work?

The Design Process, All Together 🔄

Every chapter in this guide is one piece of this cycle.



REMEMBER

Design is a loop, not a line. Team StudyBuddy went back to Empathize and Define more than once — that's not a mistake, that's the process working.

Full Glossary 📚

All 52 terms from the unit, A–Z.

- **Accessibility** — How easily a product can be used by people with different abilities.
- **App Idea** — A specific concept for an app that solves a real need for a real user.
- **App Lab** — The Code.org tool used to build real, working apps.
- **Apps for Good** — Apps designed to help solve a real problem, not just for profit.
- **Barrier** — Something that gets in the way of a user doing what they need to do.
- **Brainstorm** — Generating as many ideas as possible without judging them yet.
- **Bug** — Something in your app that doesn't work the way it's supposed to.
- **Button** — A UI element a user taps or clicks to trigger an action.
- **Category** — A group of related ideas, organized to be easier to think about.
- **Code.org / Code Studio** — The website where most hands-on project work happens.
- **Competitor** — An existing app or product that solves a similar problem to yours.
- **Criteria** — The specific standards a solution needs to meet to be successful.
- **Debug** — Finding and fixing something that isn't working correctly.
- **Define** — Clearly stating what problem you're solving and for whom.
- **Design Mode** — The part of App Lab where you build a screen's visuals before adding code.
- **Design Rationale** — The reasons behind a design decision — the WHY, not the WHAT.
- **Dropdown** — A UI element that shows a list of options when tapped.
- **Empathy** — Understanding and sharing another person's feelings or perspective.
- **Ethical** — Doing what's right and fair, especially with users and their information.
- **Evidence** — Specific facts or details that support a design decision.
- **Event** — Something a user does that tells the app to respond.
- **Feature** — Something your app does or offers, whether finished yet or not.
- **Feedback** — Specific, useful comments about how well something is working.
- **Icon** — A small picture or symbol representing an action or piece of information.
- **Inclusive** — Designed to work well for as many different kinds of people as possible.
- **Insight** — A useful realization about a user that explains why they need something.

Full Glossary (cont.) 📚

- **Interview** — Asking someone questions directly to learn about their needs or habits.
- **Iteration** — Making a design better through repeated rounds of improving it.
- **Link (Screens)** — Connecting one screen to another so tapping something moves the user.
- **Market Research** — Looking at what already exists to understand what's missing.
- **Navigation** — How a user moves between different screens or sections of an app.
- **Need** — Something a user requires from a product for it to actually work for them.
- **Opportunity** — A chance to solve a problem or make something better for a user.
- **Paper Prototype** — A version of an app made entirely out of paper, used to test an idea.
- **Perspective** — Seeing a situation from someone else's point of view.
- **Presentation** — Sharing your work and explaining your process to an audience.
- **Prioritize** — Deciding what to work on first based on what matters most.
- **Prototype** — An early, rough version of a design built to test an idea.
- **Reflection** — Thinking back on what worked, what didn't, and what you'd change.
- **Responsible** — Making design choices that consider the wellbeing and rights of users.
- **Revise** — Changing something to make it better, based on feedback.
- **Screen** — One individual view or page within an app.
- **Screen Flow** — The path a user follows moving from one screen to the next.
- **Section / Join Code** — The class-specific code students enter to join the class section.
- **Stakeholder** — Anyone who has an interest in or is affected by a project.
- **Test** — Having a real person try to use your design so you can see what works.
- **Text Input** — A UI element where a user types information into an app.
- **User** — The person a product or design is actually made for.
- **User Interface** — The screens, buttons, and controls a person sees and touches.
- **User Profile** — A description of a specific person's age, abilities, habits, and needs.
- **Usability** — How easy and pleasant something is to actually use.
- **Visual Hierarchy** — Arranging elements so the most important thing is noticed first.

Are You Ready to Present? ✅

A self-check before your final App Presentation.

- [ ] **Design Process Explanation** — Can I clearly explain the user need, my research, and how my idea changed over time?
- [ ] **Working Digital Prototype** — Are all my planned screens built and linked so someone could navigate my app easily?
- [ ] **Feedback Incorporation** — Can I point to at least one specific change I made because of real feedback?
- [ ] **Bugs/Features Handling** — Did I track my bugs/features list and fix or add the most important ones?
- [ ] **Presentation Delivery** — Can I present clearly, covering every required part, within the time given?
- [ ] **Reflection** — Do I have an honest reflection — a real strength AND a real area I'd improve?

You Started With Empathy. You're Finishing With a Real App.

From one honest conversation about a real problem, to sketches, to feedback, to bugs, to a working prototype — that's the whole design process, and you did every part of it.

"Good design today, a real app tomorrow." ♥